

Sudarshan

Final Frontend UI Component Guide

SIH 2026 Problem Statement 26227

Semantic Retrieval and Multi-Temporal Change Analysis of Satellite Imagery

Four reference screens, component-by-component explanation, and implementation intent

What this guide covers

- Search + AOI workflow
- Temporal change evidence
- False-alarm handling
- Similar sites + clustering
- Review + provenance
- Offline/local demo boundary

Primary demo query

"newly built structures near a river"

Secondary query: "large vehicle concentrations on open ground"

Dataset strategy

Target 30-38 local imagery items, grouped into four controlled cases.

Do not collect a large archive merely to populate the UI.

How the UI maps to the required capabilities

Semantic + multimodal retrieval	Search & Explore tabs, natural-language query, image search, ranked result cards, filters.
Multi-temporal change analysis	Before/After, Change Mask, Temporal Viewer, earliest supported change.
False-alarm + quality handling	Quality panel, factor-by-factor assessment, suppressed seasonal and quality-confounded cases.
Discovery + clustering	Similar Sites, Clustering, similarity scores, cluster overview.
Analyst workflow + provenance	Review Queue, Confirm/Reject, audit state, Data Provenance, Export.
Scale + incremental ingestion	Data Ingestion, Catalog, incremental update UI, local index status.
Offline + sovereignty	Persistent offline status, local archive banner, no runtime cloud dependency.

This mapping is a UI design interpretation of the supplied SIH 26227 requirements; implementation values must be verified by the backend/data pipeline.

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Component-by-component implementation notes

Use these numbers to turn the reference screen into reusable React components and future service contracts.

1

Trust header

Product identity, mission context, offline state, local archive, GPU/model status and analyst session.

2

System status cluster

Persistent offline/local indicators establish the deployment boundary for the demo.

3

Primary navigation

Moves through analyst workspace modules without breaking the current investigation.

4

Five-step workflow

Ingest -> AOI -> Index/Embed -> Cluster/Analyze -> Explore/Review; shows end-to-end process.

5

Data ingestion card

Upload/drag-and-drop zone for GeoTIFF/COG/approved local assets.

6

Staged asset list

Shows asset names, tile counts and processing state; future source of truth is the local manifest.

7

AOI controls

Draw, upload GeoJSON, or select from list; keeps spatial constraint explicit.

8

AOI map

Interactive visual target for polygon selection, site markers, river context, legend and scale.

9

AOI details panel

Displays AOI name, area, coordinates and tile count without hiding geospatial provenance.

10

Clustering controls

Select clustering method and cluster count; designed to map to an embedding/index service later.

11

Cluster legend

Explains cluster categories so colors are interpretable rather than decorative.

12

Temporal site selector

Selects the site whose local observation sequence is being reviewed.

13

Temporal image strip

Chronological observations make persistence and change progression visible.

14

Playback and timeline

Manual or play-through review of only locally staged observations.

15

Change visualization + AI insight cards

Change mask, evidence summary, first supported change and supporting metrics; must be backed by real data when integrated.

Implementation rule: preserve the visual hierarchy, but connect values to local data adapters instead of inventing production metrics.

Yellow numbers correspond to the component explanations on the next page.

Component-by-component implementation notes

Use these numbers to turn the reference screen into reusable React components and future service contracts.

1

Brand and product purpose
Shows Sudarshan plus a concise description of semantic retrieval and temporal change.

2

Organisation/trust context
Provides the institutional context present in the supplied reference design.

3

Offline status chip
Makes local/offline operation visible at all times.

4

Primary navigation
Search, change analysis, similar sites, review, catalog, analytics and settings entry points.

5

Natural-language query box
Main semantic retrieval input. This is the entry point for the exact demo query.

6

AOI mode switcher
Draw on map, upload GeoJSON, or select from list.

7

Date range control
Constrains the archive search by time.

8

Quick date presets
Speeds repeated analyst workflows while preserving a custom option.

9

Sensor selector
Keeps Sentinel-1/Sentinel-2/Landsat/Bhuvan choices explicit.

10

Additional filters
Cloud cover, region, resolution and product type refine ranked retrieval.

11

Map & AOI preview
Shows the geographic extent and context for the query.

12

Map navigation controls
Zoom, navigation and reference controls support spatial inspection.

13

Quick queries
Give judges a controlled way to trigger alternate semantic searches without typing.

14

Reset/search actions
Clear filters or execute the query; primary action must be obvious.

15

Demo boundary
The local/offline badge makes it clear the current frontend is a staged demonstration, not a live network service.

Implementation rule: preserve the visual hierarchy, but connect values to local data adapters instead of inventing production metrics.

Yellow numbers correspond to the component explanations on the next page.

Component-by-component implementation notes

Use these numbers to turn the reference screen into reusable React components and future service contracts.

1

Trust header
Same persistent identity and offline boundary.

2

Mode badge
Explicit local-demo disclosure; should disappear only after real local inference is verified.

3

Analyst session
User/session context for review and audit events.

4

Search mode tabs
Text Search, Image Search, Polygon Search and Advanced Filters.

5

Semantic query input
The exact search string stays visible so the analyst knows what was retrieved.

6

Filter chip row
AOI, date, sensor, cloud and other constraints remain auditable.

7

Ranked result list
Candidate sites ordered by retrieval relevance; each card shows semantic category and context.

8

Map view / heatmap
Links ranked results to geographic context and broader discovery.

9

Selected site panel
Shows site ID, confidence, type, coordinates, change summary and location context.

10

Before/after comparison
Core evidence view for temporal change analysis.

11

Similar-sites strip
Surfaces related locations without requiring a new query.

12

Cluster analysis
Embedding/spatial grouping view for unsupervised discovery.

13

Quality and false-alarm panel
Makes confounders explicit instead of treating every visual difference as real change.

14

Analyst tools
Add to review queue, confirm, reject, find similar sites and export.

15

Data provenance bar
Source/processing provenance entry point for the selected result.

16

Export and report actions
Generate or export evidence packages without losing provenance.

Implementation rule: preserve the visual hierarchy, but connect values to local data adapters instead of inventing production metrics.

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Component-by-component implementation notes

Use these numbers to turn the reference screen into reusable React components and future service contracts.

1

Brand header

Sudarshan identity and product purpose.

2

Institutional context

MoD / Indian Army (DGIS) trust framing as shown in the supplied design.

3

Offline/local indicators

Persistent network and local system state.

4

Search configuration column

All query, AOI, time, sensor and filter controls live in one stable workspace.

5

Search mode tabs

Text query, image search and example query pathways.

6

AOI and date controls

Spatial and temporal constraints are explicit and editable.

7

Sensor + filter section

Combines source selection with quality and context filters.

8

High-confidence toggle

Optional user control for narrowing the visible result set; must map to actual confidence logic later.

9

Search CTA

Single primary action that launches retrieval.

10

Map + split view

Results are spatially inspectable and can be compared in map/split modes.

11

Search result cards

Eight visible cards form the ranked review queue entry point.

12

Right-side site analysis header

Selected site, type, coordinates and confidence are immediately visible.

13

Before/after imagery

Direct visual evidence of the selected site across time.

14

Change detection result

Change type, confidence, earliest supported observation and change mask.

15

Temporal timeline

Tracks observations and highlights the supported change point.

16

Quality and analyst actions

False-alarm factors plus Similar Sites, Review Queue and Export Report actions.

Implementation rule: preserve the visual hierarchy, but connect values to local data adapters instead of inventing production metrics.